# Chapter 11: Brain Development and Plasticity — Mathematical Equations

*Comprehensive Neuroscience Reference*

**Abstract**

This chapter presents the mathematical foundations of developmental neuroscience and adult neuroplasticity. We derive models for neurogenesis (logistic proliferation, progenitor pool expansion), synaptogenesis and synaptic pruning (Hill kinetics, exponential elimination), critical period dynamics (parvalbumin maturation, perineuronal net accumulation), myelination progression, ocular dominance plasticity after monocular deprivation, use-dependent cortical reorganization, and epigenetic regulation (DNA methylation, histone acetylation). Allometric scaling relationships characterize brain size evolution. Six case studies with numerical solutions illustrate developmental trajectories: synapse density peaks in visual cortex at 8 months postnatal; 40–50% of synapses are eliminated from infancy to adulthood; critical period reopening via PNN degradation restores adult plasticity; myelination timing predicts functional maturation; cortical maps expand with training; and BDNF methylation links childhood trauma to adult depression risk. All equations include typical parameter values and mechanistic explanations.

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# 1. Neurogenesis and Cell Proliferation

## 1.1 Peak Neurogenesis Rate During Fetal Development

$$\frac{dN}{dt} = r_{\max} N(t) \left(1 - \frac{N(t)}{K}\right) \tag{1}$$

$$N(t) = \frac{K}{1 + e^{-r_{\max}(t - t_{50})}} \tag{2}$$

| **Symbol** | **Meaning** | **Typical Value** |
|---|---|---|
| $N(t)$ | Number of neurons produced (cumulative or rate at time $t$) | $10^5$–$10^8$ neurons |
| $r_{\max}$ | Maximum proliferation rate | 0.1–0.3 day$^{-1}$ |
| $K$ | Carrying capacity (total neurons at peak production) | ∼170 billion neurons |
| $t_{50}$ | Time of maximum production rate (inflection point) | Gestational weeks 12–20 |
| Peak rate | Neurogenesis at maximum | ∼250,000 neurons/min |
| $t$ | Gestational age | Weeks 3–25 |

**Explanation**: Neurogenesis follows a logistic (S-shaped) curve: slow initial proliferation, rapid exponential expansion in the middle trimester (peak ∼250,000 neurons/min), then plateau as progenitor pools deplete. The ventricular zone undergoes symmetric division (expanding progenitor pool) followed by asymmetric division (one progenitor + one neuron), with the switch controlled by transcription factors (Ngn2, Neuro-D). By gestational week 25, virtually all cortical neurons are born; neurogenic zones switch to producing glial cells. The logistic model captures this transition naturally: the term $(1 - N/K)$ implements saturation — as neurons accumulate, fewer progenitors remain, slowing production.

## 1.2 Progenitor Pool Expansion via Symmetric Division

$$P(t) = P_0 \times 2^{n(t)} = P_0 \times 2^{t/T_{\text{div}}} \tag{3}$$

$$n(t) = \frac{t}{T_{\text{div}}} \quad \text{(Number of division cycles)} \tag{4}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $P(t)$ | Progenitor pool size at time $t$ | $10^4$–$10^6$ cells |
| $P_0$ | Initial number of neural progenitors | ∼1000–2000 cells |
| $T_{\text{div}}$ | Division cycle time | 12–24 hours |
| $n(t)$ | Number of division cycles by time $t$ | |
| Expansion factor | $2^n$ (exponential doubling) | 10–20 cycles → $10^6$-fold |

**Explanation**: Neural progenitors initially undergo symmetric division, each cell dividing to produce two daughter progenitors. This geometric expansion (doubling per cycle) rapidly builds a large pool of proliferative cells from a small starting population. The human cortex achieves its massive size partly through extended periods of symmetric division and expanded intermediate progenitor pools (especially outer radial glia in the outer SVZ, a cell type barely present in rodents but abundant in primates). The number of symmetric cycles determines the final number of cortical neurons. Variations in division cycle number (controlled by intrinsic factors and signaling environment — FGF, SHH, Notch) explain differences in cortical size across species and individuals.

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# 2. Synaptogenesis and Synaptic Overproduction

## 2.1 Developmental Trajectory of Synapse Density

$$\rho(t) = \rho_{\max} \frac{(t - t_0)^n}{K_\rho^n + (t - t_0)^n} \quad \text{(Rise phase)} \tag{5}$$

$$\rho(t) = \rho_{\max} e^{-\lambda(t - t_{\text{peak}})} \quad \text{(Pruning phase)} \tag{6}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\rho(t)$ | Synapse density | $10^{-2}$–$10^0$ synapses/mm$^3$ |
| $\rho_{\max}$ | Peak synapse density (∼150% of adult levels) | 1.5–2.0× adult density |
| $t_0$ | Time of synaptogenesis onset | 0–4 weeks postnatal |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $t_{\text{peak}}$ | Time of maximum synapse density | 8 months (V1); 3 months (auditory); 3.5 years (PFC) |
| $K_\rho$ | Semisaturation constant (rise phase) | Reflects rate of synapse formation |
| $\lambda$ | Pruning rate constant | 0.01–0.1 month$^{-1}$ |
| $n$ | Hill coefficient (sharpness of rise) | 2–4 |

**Explanation**: Synaptogenesis shows a characteristic developmental trajectory: gradual rise postnatally (driven by axon-dendrite contact, activity, and neurotrophic factors), peak at different ages depending on cortical area, then protracted pruning. The rising phase follows a Hill equation (sigmoidal), reflecting cooperative synapse formation processes. The pruning phase follows exponential decay (monoexponential kinetics of elimination). Peak synapse density is 40–50% higher than adult steady-state, reflecting overproduction and subsequent selective elimination. The timing of peak by region correlates with the functional maturation timeline: sensory areas mature earlier (lower sensory input demands at birth); association areas, especially PFC, mature much later (complex multimodal processing and executive functions require extended development).

## 2.2 Total Synaptic Count During Development

$$S_{\text{total}}(t) = \int_V \rho(t)\, dV = V_{\text{cortex}} \times \bar{\rho}(t) \tag{7}$$

$$S_{\text{total}}(\text{peak}) \approx 1000 \text{ trillion} = 10^{15} \text{ synapses} \tag{8}$$

$$S_{\text{total}}(\text{adult}) \approx 100 \text{ trillion} \approx 10^{14} \text{ synapses} \tag{9}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $S_{\text{total}}(t)$ | Total synaptic count in cortex at time $t$ | Synapses |
| $V_{\text{cortex}}$ | Cortical volume | $\sim$1000 cm$^3$ (adult) |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\bar{\rho}(t)$ | Average synapse density across cortex | Increases with development |
| At peak (infancy) | ∼1,000 trillion synapses in infant cortex | 10-fold more than adult |
| At adulthood | ∼100 trillion synapses in adult cortex | Pruned to 10% of peak |
| Pruning fraction | 40–50% of peak synapses eliminated | Refines circuits, reduces noise |

**Explanation**: The infant brain contains approximately 1,000 trillion synapses — far more than needed. This synaptic exuberance serves multiple functions: (1) generates diverse neural circuits from which selective ones are retained; (2) ensures coverage and redundancy, protecting against damage; (3) provides raw material for experience-dependent specialization. Adult refinement (pruning to ∼100 trillion synapses) involves activity-dependent selection (Hebbian competition) and molecular tagging/removal (complement-mediated microglial phagocytosis). The elimination is not random — weak/uncorrelated synapses are preferentially removed, while strong/correlated ones are retained.

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# 3. Synaptic Pruning and Activity-Dependent Elimination

## 3.1 Hebbian Synaptic Plasticity

$$\Delta w = \eta \times (r_{\text{post}} - \bar{r}_{\text{post}}) \times (r_{\text{pre}} - \bar{r}_{\text{pre}}) \tag{10}$$

$$\Delta w = \begin{cases} +\alpha & \text{if } r_{\text{pre}} \text{ and } r_{\text{post}} \text{ correlated (LTP)} \\ -\beta & \text{if uncorrelated or anti-correlated (LTD)} \\ 0 & \text{if only one is active} \end{cases} \tag{11}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\Delta w$ | Change in synaptic weight | Positive (LTP) or negative (LTD) |
| $\eta$ | Hebbian learning rate | 0.01–0.1 |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $r_{\text{pre}}, r_{\text{post}}$ | Presynaptic and postsynaptic firing rates | spikes/s |
| $\bar{r}_{\text{pre}}, \bar{r}_{\text{post}}$ | Average (baseline) firing rates | spikes/s |
| $\alpha, \beta$ | LTP and LTD magnitude constants | 0.1–0.5 |
| STDP | Spike-timing-dependent plasticity | $\Delta t = \pm 20$ ms |

**Explanation**: The Hebb rule ("neurons that fire together, wire together") provides the algorithm for competition: synapses carrying correlated signals between presynaptic and postsynaptic neurons are strengthened (LTP); uncorrelated synapses weaken (LTD) and are eventually eliminated (pruned). This implements a form of error correction in sensory circuits: during critical periods, correlations in sensory input drive the formation of feedforward circuits (e.g., retinotopic maps in visual cortex). Synapses that reliably signal information (correlated with sensory structure or postsynaptic activity) strengthen; noise-carrying synapses (uncorrelated) weaken. Spike-timing-dependent plasticity (STDP) refines this: the precise timing between presynaptic spike and postsynaptic action potential determines direction (causal pre→post drives LTP; backward post→pre drives LTD). This temporal specificity allows circuits to learn predictive relationships and causal structure.

## 3.2 Complement-Mediated Synaptic Tagging and Microglial Pruning

$$C3_{\text{deposition}}(\text{synapse}) \propto \text{Activity}_{\text{low}} \times \text{Maturity}_{\text{weak}} \tag{12}$$

$$P(\text{synapse eliminated by microglia}) = \frac{C3_{\text{level}}}{\text{CR3 threshold}} \times \phi(\text{microglial activation}) \tag{13}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $C3$ | Complement protein C3 (from astrocytes, neurons) | High on inactive/weak synapses |
| $C1q$ | Upstream complement initiator protein | Elevated during pruning period |
| CR3 | Complement receptor 3 on microglia | Expression increases during pruning |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\text{Activity}_{\text{low}}$ | Synaptic activity | Correlated with synaptic strength |
| $\text{Maturity}_{\text{weak}}$ | Synapse maturity | Proxy for synapse refinement status |
| $\phi(\text{activation})$ | Microglial activation state | 0 (resting) to 1 (activated) |

**Explanation**: Beyond Hebbian competition, molecular machinery tags and physically removes weak synapses. Astrocytes and the postsynaptic cell secrete complement proteins (C1q, C3); these accumulate preferentially on weak/inactive synapses. Microglia (immune cells of the CNS) express complement receptors (CR3, CR1) that recognize C3-tagged synapses. Upon engagement, microglia are activated to engulf and phagocytose the tagged synapse. Dysregulation of this pruning is implicated in neurodevelopmental disorders: excessive pruning (high C4 expression in schizophrenia) may cause cognitive deficits; insufficient pruning (in autism or intellectual disability) may impair circuit refinement.

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# 4. Critical Period Dynamics and Plasticity Windows

## 4.1 Critical Period Opening — PV+ Maturation

$$\text{Plasticity} = f(t) = \frac{1}{1 + e^{-\beta(t - t_{\text{open}})}} \times \left(1 - \frac{t - t_{\text{open}}}{t_{\text{duration}}}\right)^{+} \tag{14}$$

$$\text{PV+ density}(t) = \text{PV}_{\text{max}} \left(1 - e^{-\lambda(t - t_0)}\right) \tag{15}$$

$$\text{Inhibitory conductance} = g_{\text{max}} \times \text{PV+ density}(t) \tag{16}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\text{Plasticity}(t)$ | Degree of experience-dependent modification | 0–1 scale |
| $t_{\text{open}}$ | Time of critical period opening | Postnatal weeks 2–4 |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $t_{\text{duration}}$ | Duration of critical period | Weeks to months |
| $\beta$ | Steepness of opening transition | 0.5–1.0 |
| PV+ density | Density of parvalbumin-positive interneurons | Increases 2–3× |
| $\lambda$ | Rate constant for PV+ maturation | 0.1–0.3 week$^{-1}$ |
| $g_{\max}$ | Maximum GABAergic conductance | Proportional to PV+ density |

**Explanation**: Critical periods open when the brain achieves sufficient neuronal maturity and circuit organization to benefit from experience. A key trigger is the maturation of parvalbumin-positive (PV+) fast-spiking interneurons, which provide powerful GABAergic inhibition. The increase in PV+ density and inhibitory drive sharpens the signal-to-noise ratio in sensory cortices: synchronous (signal-carrying) input triggers postsynaptic spiking reliably despite increased background inhibition; weak uncorrelated inputs are more effectively suppressed. This enhanced discriminability creates a window of opportunity for activity-dependent learning — experience can more effectively shape circuits because strong, structured input produces larger changes in postsynaptic responses.

## 4.2 Critical Period Closure — PNN and Myelin Brakes

$$\text{Plasticity}_{\text{closure}} = \text{Plasticity}_{\max} \times \frac{1}{1 + ([\text{PNN}]/[\text{PNN}]_{50})^2} \tag{17}$$

$$[\text{PNN}](t) = [\text{PNN}]_{\max}\left(1 - e^{-k_{\text{PNN}}(t - t_{\text{close}})}\right) \tag{18}$$

$$\text{Nogo signaling} = k_{\text{Nogo}} \times [\text{myelin}] \quad \text{(Via NgR receptors)} \tag{19}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| PNN | Perineuronal net | Low early; accumulates with age |
| $[\text{PNN}]_{50}$ | PNN for half-maximal plasticity suppression | Ref. threshold |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\text{Plasticity}_{\text{closure}}$ | Remaining plasticity after closure signals | Decreases to 0.1–0.3 |
| $k_{\text{PNN}}$ | Rate constant for PNN accumulation | 0.05–0.2 $\text{week}^{-1}$ |
| $t_{\text{close}}$ | Time of critical period closure | 6–8 weeks (mouse); 4–7 years (human) |

**Explanation**: Critical periods close when molecular "brakes" accumulate around PV+ neurons and in white matter. Perineuronal nets (PNNs) are lattice-like structures of extracellular matrix that ensheath PV+ cell soma and proximal axons. PNNs restrict the rearrangement of synaptic connections — they physically limit growth cone motility and synapse formation. Additionally, myelin-associated inhibitors (particularly Nogo, through Nogo receptor/NgR) restrict axonal growth and dendritic spine plasticity. These mechanisms work in concert to "lock in" the circuit configuration established during the critical period. The closure is gradual (not abrupt), with plasticity declining progressively from peak → residual level. Crucially, these brakes can be experimentally removed, reopening plasticity: chondroitinase ABC (enzyme that degrades PNNs), dark-rearing (reduces myelin formation), fluoxetine (enhances GABA inhibition), and HDAC inhibitors (enhance histone acetylation). This has therapeutic implications for adult amblyopia and stroke recovery.

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# 5. Myelination and White Matter Development

## 5.1 Spatiotemporal Progression of Myelination

$$M(t, \text{tract}) = M_{\max} \frac{(t - t_{\text{onset}})^n}{K_M^n + (t - t_{\text{onset}})^n} \tag{20}$$

| Symbol | Meaning | Human Timeline |
|---|---|---|
| $M(t)$ | Extent of myelination (0–1) | |
| $t_{\text{onset}}$ | Age of myelination onset | Varies by tract |
| $M_{\max}$ | Maximum myelination (adult level) | 1.0 |
| $K_M$ | Semisaturation (rate parameter) | 0.3–0.7 |
| $n$ | Hill coefficient | 2–3 |

| **Symbol** | **Meaning** | **Human Timeline** |
|---|---|---|
| Brainstem | Onset prenatal; Complete 3–6 months | Early |
| Sensory/motor | Onset term; Complete 6–12 months | Early postnatal |
| Internal capsule | Onset 6 months; Complete 2–3 years | Childhood |
| Corpus callosum | Onset 6 months; Complete 5–10 years | Mid-childhood |
| Association areas | Onset 2–3 years; Complete 10–20 years | Adolescence |
| PFC white matter | Onset 5–10 years; Complete 25–30 years | Latest to mature |

**Explanation**: Myelination progresses in a stereotyped sequence: earliest in brainstem (critical for basic functions); early in primary sensory and motor tracts (needed for sensation and motor control); latest in association cortices and PFC (which subserve complex cognition). The temporal progression reflects the functional demands: structures needed early for survival and basic functioning myelinate first; higher-order cognitive circuits myelinate last. Prefrontal white matter completes myelination in the mid-20s in humans — one of the last cortical regions to mature structurally. This late maturation underlies the protracted development of executive function, impulse control, and decision-making in adolescence.

## 5.2 Activity-Dependent Myelin Formation

$$\frac{d[\text{Myelin}]}{dt} = k_{\text{formation}} \times [\text{Oligodendrocytes}] \times \text{Activity}(t) - k_{\text{degrade}} \times [\text{Myelin}] \tag{21}$$

$$[\text{Oligodendrocytes}] = \text{OPC baseline} + k_{\text{diff}} \times \text{Experience-driven signals} \tag{22}$$

| **Symbol** | **Meaning** | **Typical Value** |
|---|---|---|
| Myelin | Myelin sheath thickness | Increases with experience |
| $k_{\text{formation}}$ | Rate constant for myelin formation | 0.1–0.5 day$^{-1}$ |

| Symbol | Meaning | Typical Value |
|---|---|---|
| Oligodendrocytes | Mature myelin-producing glial cells | ∼5–10 per axon segment |
| Activity($t$) | Neural activity in the axon | 0–1 scale |
| $k_{\text{degrade}}$ | Myelin degradation rate | 0.01–0.05 day$^{-1}$ |
| OPC | Oligodendrocyte precursor cells | ∼20–40% of white matter cells |
| $k_{\text{diff}}$ | Differentiation rate constant | Increases with learning/practice |

**Explanation**: Myelination is not just a developmental program but an ongoing, activity-dependent process. Intensely used neural pathways recruit oligodendrocyte precursor cells (OPCs), which differentiate into mature oligodendrocytes that wrap myelin around axons. This adaptive myelination increases conduction velocity (axon diameter × myelination) of frequently-used pathways, enhancing neural efficiency. Learning tasks (motor training, cognitive tasks, navigation) induce localized myelin formation in task-relevant white matter tracts, visible within weeks in rodent models. This activity-dependent regulation allows the brain to dynamically optimize communication speed based on functional demands: pathways critical for newly-learned skills receive more myelin, enabling faster, more reliable signal transmission.

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# 6. Ocular Dominance Plasticity and Visual Critical Period

## 6.1 Ocular Dominance Shift After Monocular Deprivation

$$\text{OD}_{\text{contralateral}}(t) = \text{OD}_{\text{baseline}} + \Delta_{\max}\left(1 - e^{-\lambda_{\text{MD}} \times t}\right) \tag{23}$$

$$\Delta_{\max} = (g_{\text{deprive}} - g_{\text{open}}) \times \text{Plasticity}_{\text{window}}(t) \tag{24}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\text{OD}_{\text{contralateral}}$ | Ocular dominance: fraction driven by contralateral eye | 0–1 |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\mathrm{OD}_{\mathrm{baseline}}$ | Initial ocular dominance | 0.55–0.65 |
| $\Delta_{\max}$ | Maximum OD shift toward open eye | 0.2–0.4 |
| $\lambda_{\mathrm{MD}}$ | Time constant for OD shift | 0.1–0.3 $\mathrm{day}^{-1}$ |
| $t$ | Duration of monocular deprivation | Days to months |
| $g_{\mathrm{deprive}}$ | Gain (response strength) for deprived eye | Decreases with deprivation |
| $g_{\mathrm{open}}$ | Gain for open eye | Normal or enhanced |
| Amblyopia threshold | OD shift >0.2 can cause vision loss | Needs intervention |

**Explanation**: Monocular deprivation (MD) during the critical period causes a dramatic shift in ocular dominance: cortical neurons become less responsive to the deprived eye and more responsive to the open eye. The shift reflects both a loss of deprived-eye synapses (pruning due to lack of activity/reward) and a competitive expansion of open-eye inputs. The open eye generates vigorous, structured input → drives strong postsynaptic responses → triggers Hebbian strengthening of open-eye synapses via NMDA-dependent LTP. The deprived eye generates little input → weak postsynaptic activation → fails to trigger LTP, synapses weaken, and are pruned. This plasticity is mediated by the same mechanisms as critical period opening (PV+ maturation, inhibitory sharpening) and closure (PNN accumulation, myelin brakes). The plasticity is time-limited: if deprivation occurs outside the critical period (adult), OD shifts are minimal or absent — the circuit is "locked in."

## 6.2 Critical Period Closure and PNN-Dependent Plasticity Restriction

$$\text{Recovery from MD} = \text{Recovery}_{\max} \times \frac{[\mathrm{PNN}]_{\mathrm{baseline}}}{[\mathrm{PNN}](t)} \times \text{Plasticity}_{\mathrm{closure}}([\mathrm{PNN}]) \tag{25}$$

$$[\mathrm{PNN}](t) = [\mathrm{PNN}]_0 e^{k_{\mathrm{PNN}} t} \tag{26}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| Recovery | Ability to recover (reverse) OD shift with retraining | High in critical period |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $\text{Recovery}_{\text{max}}$ | Maximum possible recovery (critical period) | 0.8–1.0 |
| PNN | Perineuronal net density | PNN> Increases exponentially |
| $k_{\text{PNN}}$ | PNN accumulation rate | 0.1–0.3 $\text{year}^{-1}$ |
| During critical period | MD reversal recovery within 1–2 weeks | ∼80% recovery |
| After critical period | Same reversal with extended retraining | <10% recovery |
| With PNN removal | Adult + chondroitinase + retraining | 50–70% recovery |

**Explanation**: The restriction in adult plasticity is explained by PNN accumulation around PV+ inhibitory interneurons. During the critical period, PNNs are sparse or absent, allowing dynamic rewiring. As the animal ages, PNNs accumulate, physically constraining synaptic rearrangement and restricting plasticity. Adult visual cortex retains some residual plasticity (partial reversal of OD shift with extensive retraining, ∼10%), but it is orders of magnitude slower and smaller than critical period plasticity. Strikingly, experimental removal of PNNs (via chondroitinase ABC) in adult animals partially restores critical period-like plasticity: animals can again achieve substantial OD recovery. This demonstrates that PNNs are a physical brake on plasticity, and their removal can effectively "reopen" the critical period. This has profound implications for treating adult amblyopia and other developmental disorders.

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# 7. Use-Dependent Cortical Reorganization

## 7.1 Representational Expansion from Training

$$A_{\text{rep}}(\text{trained feature}) = A_{\text{baseline}} \times (1 + e^{\lambda_{\text{train}} \times T}) \tag{27}$$

$$\text{Expansion factor} = \frac{A_{\text{rep}}(\text{post-training})}{A_{\text{rep}}(\text{pre-training})} = 1 + \beta \times \text{Task demand} \tag{28}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $A_{\text{rep}}$ | Areal representation (cortical area devoted to a feature) | $\text{mm}^2$ of cortex |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $A_{\text{baseline}}$ | Pre-training representation size | Reference value (1.0) |
| $T$ | Duration of training/practice | Hours to months |
| $\lambda_{\text{train}}$ | Training-induced growth rate constant | 0.01–0.1 $\text{hour}^{-1}$ |
| $\beta$ | Sensitivity to task demand | 0.1–0.5 |
| String player | 2–3× expansion of fingering hand in S1 | vs. right-hand fingers |
| Braille reader | 2–3× expansion + visual cortex recruitment | Cross-modal plasticity |

**Explanation**: Intensive use of a body part or sensory pathway causes the cortical representation to expand — the region of cortex devoted to that feature grows, reflecting more neurons tuned to that stimulus. This use-dependent expansion occurs via multiple mechanisms: (1) unmasking of existing silent synapses through Hebbian strengthening; (2) axonal sprouting and formation of new synapses; (3) changes in dendritic structure; (4) activity-dependent myelination of relevant pathways. The expansion is proportional to the amount of task-related activity and practice. Professional musicians, Braille readers, and athletes show enlarged representations of task-relevant body parts and sensory inputs. The expansion reflects a principle of neural economy: neural real estate is allocated according to behavioral demand.

## 7.2 Cross-Modal Plasticity After Sensory Deprivation

$$A_{\text{V1}}(\text{recruited for other modalities}) = A_{\text{baseline}} \times f(\text{deprivation}, t, \text{age}) \tag{29}$$

$$f(\text{deprivation}, t, \text{age}) = \begin{cases} \text{High} & \text{if early deprivation (congenital/early blind)} \\ \text{Medium} & \text{if adult deprivation, early training} \\ \text{Low} & \text{if adult deprivation, no training} \end{cases} \tag{30}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $A_{\text{V1}}$ | Activation of visual cortex during non-visual tasks | fMRI BOLD signal |
| Deprivation | Sensory loss (blindness, deafness) | Age-dependent plasticity |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $t$ | Time post-deprivation | Days to years |
| Age | Age at deprivation onset | Early (<5 yrs) vs. adult |
| Early blind | Congenital/early childhood blindness | High cross-modal plasticity |
| Late-blind | Adult onset blindness | Reduced plasticity |

**Explanation**: The brain is fundamentally flexible — when a sensory modality is lost, the deprived sensory cortex does not remain idle; instead, it is recruited to process other modalities. This cross-modal plasticity is dramatic in early-blind individuals: visual cortex (V1, V2) activates during Braille reading (tactile), sound localization (auditory), and language tasks (linguistic). Moreover, these activations are functionally significant: TMS to V1 in early-blind Braille readers disrupts Braille reading performance, proving that V1 contributes to the task. The plasticity is strongest with early (congenital/early childhood) deprivation and intensive compensatory use of other senses; late-onset (adult) deprivation shows much less cross-modal recruitment.

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# 8. Epigenetic Gene Regulation

## 8.1 DNA Methylation and Gene Expression

$$P(\text{gene expressed}) = 1 - m(\text{promoter}) \times f_{\text{silencing}} \tag{31}$$

$$\text{Methylation level} = \frac{[\text{Methylated cytosine}]}{[\text{Total cytosine}]} \times 100\% \tag{32}$$

$$[\text{Methylated cytosine}](t) = [\text{Methyl}]_{\text{max}}(1 - e^{-k_{\text{meth}} \cdot t}) \tag{33}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| $m$ | DNA methylation level at gene promoter | 0–1 |
| CpG | Cytosine-guanine dinucleotide (target site) | ∼25,000 CpG sites per cell |
| $f_{\text{silencing}}$ | Silencing efficiency | 0.5–1.0 |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $k_{\text{meth}}$ | Methylation rate constant | 0.01–0.1 $\text{day}^{-1}$ |
| BDNF promoter | High methylation in depressed individuals | Trauma → methylation |
| FKBP5 gene | Childhood trauma + risk genotype → demethylation | Lifelong HPA dysregulation |

**Explanation**: DNA methylation is a major epigenetic mechanism controlling gene expression. Methyl groups ($-CH_3$) added to cytosines in CpG dinucleotides (promoter regions especially) block transcription factor binding and recruit silencing machinery (methyl-binding proteins), suppressing gene expression. Conversely, promoter unmethylation permits transcription factor access and gene activation. Developmental experience and stress regulate methylation patterns: childhood adversity increases methylation of genes encoding BDNF and related plasticity factors, reducing their expression and impairing resilience. Conversely, enriched experience and learning promote demethylation of synaptic plasticity genes (via TET enzyme activity), enhancing their expression and supporting learning. This provides a molecular mechanism linking early environment to long-term brain function and mental health outcomes.

## 8.2 Histone Acetylation and Chromatin Remodeling

$$\text{Transcription rate} = k_0 \times \frac{[\text{Acetyl-histone}]}{[\text{Acetyl-histone}]_{50}} \times P(\text{TF binding}) \tag{34}$$

$$\frac{d[\text{Acetyl-histone}]}{dt} = k_{\text{HAT}} - k_{\text{HDAC}} \times [\text{Acetyl-histone}] \tag{35}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| Acetyl-histone | Histone with acetyl groups on lysine residues | Increases during learning |
| HAT | Histone acetyltransferase (adds acetyl groups) | CREB-binding protein (CBP) |
| HDAC | Histone deacetylase (removes acetyl groups) | 18 mammalian HDACs |
| Open chromatin (euchromatin) | Acetylated histones, transcription active | DNA accessible to TFs |

| Symbol | Meaning | Typical Value |
|---|---|---|
| Closed chromatin (heterochromatin) | Deacetylated histones, transcription silent | Genes silenced |
| $k_{\text{HAT}}, k_{\text{HDAC}}$ | Rate constants | 0.1–1.0 per minute |

**Explanation**: Histone acetylation is a complementary epigenetic mechanism to DNA methylation. Histones are the protein scaffolds around which DNA wraps; acetylation of histone lysines neutralizes positive charges → loosens DNA-histone contacts → "opens" chromatin, making DNA accessible to transcription factors. This elevated histone acetylation at plasticity genes (c-fos, BDNF, CREB-target genes) during learning reflects active transcription. Conversely, deacetylation (via HDAC enzymes) closes chromatin, silencing genes. Learning experiences trigger transient increases in HAT activity and histone acetylation at memory-relevant genes, enabling rapid transcription of proteins needed for memory consolidation. Remarkably, HDAC inhibitors (which prevent deacetylation, maintaining acetylation) enhance memory formation and can reopen developmental critical periods, suggesting that chromatin remodeling is a key control point for plasticity.

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# 9. Brain Size Evolution and Encephalization

## 9.1 Allometric Scaling (Encephalization Quotient)

$$\text{Brain Volume} = a \times (\text{Body Mass})^b \tag{36}$$

$$\text{Encephalization Quotient (EQ)} = \frac{\text{Actual Brain Volume}}{\text{Expected Brain Volume (for body size)}} = \frac{B}{aM^b} \tag{37}$$

| Symbol | Meaning | Typical Value |
|---|---|---|
| Brain Volume | Total brain volume or mass | $cm^3$ or grams |
| Body Mass | Body mass of organism | kg |
| $a$ | Allometric intercept | 0.1–1.0 |
| $b$ | Allometric exponent | ∼0.67 |
| EQ | Encephalization quotient | 1.0 = average |

| **Symbol** | **Meaning** | **Typical Value** |
|---|---|---|
| Humans | EQ ∼7 (7× larger than expected) | Largest EQ among primates |
| Great apes (chimp) | EQ ∼2–2.5 | Significantly larger-brained |
| Old World monkeys | EQ ∼2–3 | Larger-brained than expected |
| Ungulates | EQ ∼0.8–1.2 | Near-average for mammals |
| Rodents | EQ ∼0.5–1.0 | Smaller-brained than primates |

**Explanation**: Brain size scales with body size across species, but not isometrically (not a 1:1 relationship). Instead, brain volume scales with body mass to the 2/3 power (or ∼0.67 exponent). This means larger-bodied animals have proportionally smaller brains. The encephalization quotient (EQ) normalizes for body size, revealing which species are "over-brained" (high EQ: humans, dolphins, elephants) vs. "under-brained" (low EQ: most fish). Humans have the highest EQ among primates (∼7) and among all mammals (∼7, competing with elephants and dolphins for top spot). The human brain occupies ∼2% of body mass but consumes ∼20% of energy — an enormous metabolic cost, suggesting strong selective pressure for intelligence/cognitive sophistication.

## 9.2 Human-Specific Brain Expansion — Gene Effects on Progenitors

$$\text{Cortical neurons} = P_0 \times 2^{n_{\text{divisions}}} \quad \text{(Progenitor expansion)} \tag{38}$$

$$\text{Layer composition} = \int_0^t \text{Neurogenesis rate}(t')\,dt' \quad \text{(Integration of birth times)} \tag{39}$$

$$\text{Expansion factor (Human vs. Chimpanzee)} = \frac{[\text{Gene}_{\text{human}}]}{[\text{Gene}_{\text{chimp}}]} \times \text{Effect size} \tag{40}$$

| **Symbol** | **Meaning** | **Typical Value** |
|---|---|---|
| $P_0$ | Initial number of neural progenitors | 1,000–10,000 cells |

| Symbol | Meaning | Typical Value |
|---|---|---|
| $n_{\text{divisions}}$ | Number of division cycles | 15–25 cycles |
| Human advantage | Expanded outer SVZ → more neurons | 2–3× more than chimps |
| ASPM, MCPH1, CDK5RAP2 | Centrosomal proteins in progenitor divisions | Positive selection in humans |
| ARHGAP11B | Human-specific gene duplication | ∼10% increase in outer SVZ |
| SRGAP2C | Human-specific duplication | Promotes spine density |
| HAR1 | Non-coding RNA; Cajal-Retzius cells | Rapidly evolved |
| FOXP2 | Transcription factor; language/speech | Human-specific amino acid changes |
| Effect of ARKGAP11B in mice | Ectopic expression → cortical folding | Normally smooth-brained |

**Explanation**: The human brain's massive size compared to our evolutionary relatives (particularly chimp) results from genetic and developmental changes that expand the progenitor pool and extend neurogenesis duration. Key genes implicated in human brain expansion (ASPM, MCPH1, CDK5RAP2, ARHGAP11B, SRGAP2C) control progenitor cell division and migration. These genes underwent positive selection in the human lineage — meaning mutations that increased their activity or expression were favored by evolution. ARHGAP11B is particularly striking: it's a human-specific gene duplication (found only in humans, not other apes) that emerged ∼5 million years ago. When expressed in mice (which normally lack it), it causes their cortex to fold — something that never happens in wild mice. This demonstrates that a single human-specific gene is sufficient to partially replicate human-like cortical expansion. The mechanisms involve increasing the number of intermediate progenitors in the outer SVZ, extending symmetric divisions, and delaying neuronal differentiation — all of which amplify the final number of neurons.

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## 10. Case Studies with Numerical Solutions

## 10.1 Case Study 1: Synapse Density Development — Visual Cortex Peak at 8 Months

**Scenario**: Model the developmental trajectory of synapse density in human visual cortex, from birth through adulthood.

**Given**:

- Synapse density at birth: $\rho(0) = 0.3 \times \rho_{\text{max}}$
- Peak synapse density: $\rho_{\text{max}} = 1.5 \times 1.0$ (normalized)
- Time of peak: $t_{\text{peak}} = 8$ months
- Adult steady-state: $\rho_{\text{adult}} = 1.0$ (normalized)
- Rise phase Hill parameters: $n = 2.5$, $K_\rho = 0.3$
- Pruning rate constant: $\lambda = 0.05$ month$^{-1}$

**Find**:

1. Synapse density at 3, 6, 8, 12 months, 2 years, 10 years
2. Total number of synapses in visual cortex at each timepoint
3. Rate of synapse formation and elimination at key stages

**Solution**:

Rise phase ($t = 0$ to $t_{\text{peak}} = 8$ months):

$$\rho(t) = 1.5 \frac{(t/8)^{2.5}}{0.3^{2.5} + (t/8)^{2.5}}$$

**At 3 months** ($t = 3$):

$$\rho(3) = 1.5 \frac{(0.375)^{2.5}}{0.027 + (0.375)^{2.5}} = 1.5 \times 0.53 = 0.80$$

**At 6 months** ($t = 6$):

$$\rho(6) = 1.5 \frac{(0.75)^{2.5}}{0.027 + (0.75)^{2.5}} = 1.5 \times 0.94 = 1.41$$

**At 8 months (peak)**: $\rho(8) = 1.5$ (by definition)

Pruning phase ($t > 8$ months): $\rho(t) = 1.5 \times e^{-0.05(t-8)}$

**At 12 months** ($t = 12$; 4 months into pruning):

$$\rho(12) = 1.5 \times e^{-0.20} = 1.5 \times 0.82 = 1.23$$

**At 2 years** ($t = 24$ months; 16 months of pruning):

$$\rho(24) = 1.5 \times e^{-0.80} = 1.5 \times 0.45 = 0.68$$

**At 10 years** ($t = 120$ months; 112 months of pruning):

$$\rho(120) = 1.5 \times e^{-5.6} \approx 1.0$$

**Summary Table**:

| Age | $\rho(t)$ (normalized) | % of peak | % of adult |
|---|---|---|---|
| Birth | 0.30 | 20% | 30% |
| 3 months | 0.80 | 53% | 80% |
| 6 months | 1.41 | 94% | 141% |
| **8 months (PEAK)** | **1.50** | **100%** | **150%** |
| 12 months | 1.23 | 82% | 123% |
| 2 years | 0.68 | 45% | 68% |
| 5 years | ∼1.05 | 70% | 105% |
| 10 years | 1.00 | 67% | 100% |

Assuming human visual cortex volume ∼10 cm$^3$ and adult synapse density ∼1 synapse/$\mu$m$^3$ (∼5×10$^{13}$ synapses in adult V1):

| Age | Synapses in V1 (×10$^{13}$) | % of adult |
|---|---|---|
| Birth | 1.5 | 30% |
| 3 months | 4.0 | 80% |
| 8 months (PEAK) | 7.5 | 150% |
| 12 months | 6.2 | 123% |
| 2 years | 3.4 | 68% |
| 10 years | 5.0 | 100% |

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## 10.2 Case Study 2: Synaptic Pruning Over Childhood — 40–50% Elimination

**Scenario**: Calculate the cumulative fraction of synapses eliminated from peak density to adulthood.

**Given**:

- Peak synaptic count: $S_{\text{peak}} = 1.0 \times 10^{15}$ (entire cortex)
- Adult synaptic count: $S_{\text{adult}} = 1.0 \times 10^{14}$ (entire cortex)

**Overall pruning**:

$$\text{Pruning fraction} = \frac{S_{\text{peak}} - S_{\text{adult}}}{S_{\text{peak}}} = 0.90 = 90\%$$

**Regional pruning rates**:

| Region | Peak age | Adult (% of peak) | Pruning | Timeline |
|---|---|---|---|---|
| V1 | 8 months | 50–60% | 40–50% | Birth–5 yrs; 5–10 yrs (gradual) |
| Auditory | 3 months | 60% | 40% | Birth–3 yrs |
| Motor | 6 months | 60% | 40% | Birth–5 yrs |
| PFC | 3–5 years | 50–60% | 40–50% | 5–15 yrs (protracted) |
| Global | — | 10% | ∼90% | Entire developmental period |

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